

EASY BUILD: TECHNOLOGY ARCHITECTURE BLUEPRINT

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1. System Overview & Core Databases

The technology infrastructure at EASY BUILD is built around a microservices architecture hosted on AWS. Key databases include a PostgreSQL instance that stores the ERP core and transactional order management schemas, and a Redis cluster for session caching.

Refer to the system architecture diagram below for service endpoints, API gateway routing, and database communication ports.

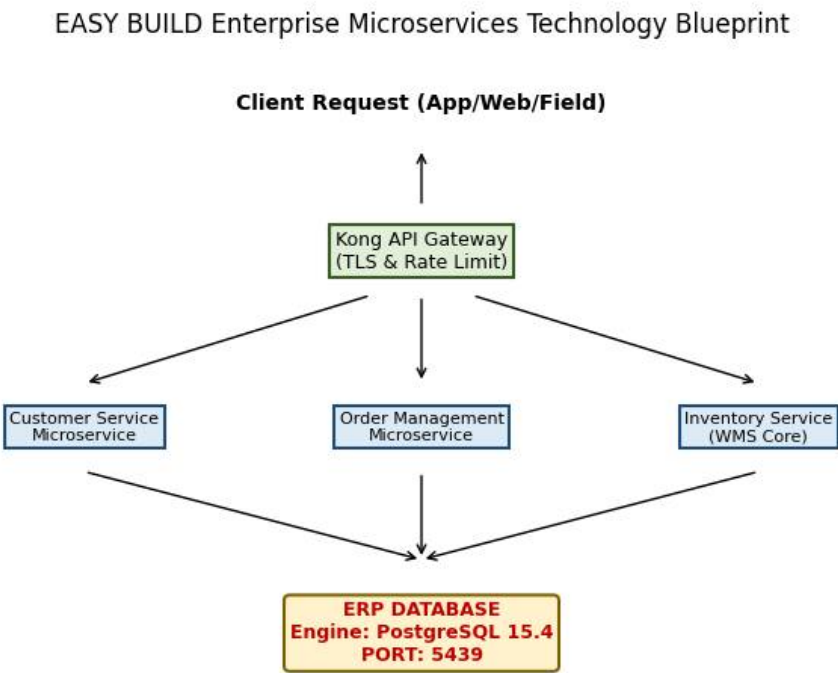


Figure 9.1: Microservices deployment architecture.

Vector embeddings are generated using local transformer models and stored in ChromaDB vector stores with cosine similarity configurations, allowing fast semantic similarity search.

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2. API Gateway and Communication Protocol

Microservices communicate asynchronously via Kafka event streams and synchronously via REST APIs routed through Kong Gateway Enterprise. OAuth 2.0 and JWT tokens secure all client interactions.

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3. WMS and ERP Integrations

The Warehouse Management System (WMS) is integrated in real-time with the central ERP database. This ensures that inventory checks and bin allocations are synced instantly, preventing double allocation of stock.

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In this context, API schemas must follow OpenAPI specifications. Every service must support health check endpoints, structured JSON logging, and distributed tracing metadata headers (X-Request-ID).

Moreover, the Order Management service interacts with the ERP core database over port 5439. The database runs on PostgreSQL 15.4 to handle complex ACID transactions and structured JSON metadata queries.

Additionally, the Order Management service interacts with the ERP core database over port 5439. The database runs on PostgreSQL 15.4 to handle complex ACID transactions and structured JSON metadata queries.

In this context, vector embeddings are generated using local transformer models and stored in ChromaDB vector stores with cosine similarity configurations, allowing fast semantic similarity search.

Additionally, the microservices architecture employs Kong API Gateway for routing client calls, securing endpoints via OAuth 2.0, and applying rate limits to protect backend services from denial of service.